

DrepaViT: An interpretable hybrid CNN–Transformer model for improved red blood cell classification

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Abstract.

We present DrepaViT, a novel hybrid deep learning model for red blood cell classification in the context of sickle cell disease (SCD) diagnosis. Combining ResNet-50 and Vision Transformer (ViT) architectures, DrepaViT captures both local and global morphological features. Trained on the SKDrepaData-v2 dataset—a curated collection of 1,489 annotated microscopy images—our model achieves state-of-the-art performance with 93.33% accuracy, 93.30% F1-score, and an AUC of 0.9456. We further enhance interpretability using Grad-CAM, demonstrating clinically relevant feature localization. A web-based interface supports real-time analysis, improving diagnostic accessibility in low-resource settings. Our results highlight DrepaViT’s potential as a robust, interpretable tool for automated hematological screening of SCD.

Keywords: Sickle Cell Disease, Red Blood Cell Classification, Vision Transformer, Deep Learning, Explainable AI, Medical Imaging.

1 Introduction

Sickle cell disease (SCD) is a severe genetic disorder caused by a point mutation in the β -globin gene, resulting in abnormal hemoglobin S (HbS) [1]. Under low oxygen, HbS polymerizes, deforming red blood cells into a sickle shape and causing vaso-occlusion, anemia, and serious complications [2], [3]. SCD affects over 300,000 newborns annually in sub-Saharan Africa [4]; in Senegal, 1.9% of newborns present the severe SS form, and 11.1% of the population carries the HbS allele [5], [6], [7]. Without early screening, more than half of affected children die before age five.

Traditional diagnosis relies on manual microscopy, limited by subjectivity, cost, and the need for skilled personnel—resources often lacking in low-income regions [8], [9], [10]. These constraints hinder early detection, while computer-aided diagnosis (CAD) systems offer a scalable solution through automated red blood cell classification [11], [12].

Although convolutional neural networks (CNNs) have achieved strong results in medical imaging, their local receptive fields limit their ability to model global dependencies—critical in SCD due to the high morphological variability of sickled cells and

varying smear conditions[13], [14]. Vision Transformers (ViTs) have emerged as a promising solution, leveraging self-attention to model long-range spatial relationships [15]. However, ViTs require large annotated datasets—often impractical in clinical settings due to the cost and scarcity of expert-labelled data. This has driven the development of hybrid architectures that combine CNNs and Transformers to capture both local and global features for robust SCD detection.

In this study, we introduce DrepaVit, a novel hybrid model that combines ResNet-50 [16] for local morphological feature extraction with Vision Transformers (ViT) [15] for global contextual modelling. Designed for enhanced diagnostic accuracy and reduced processing time, DrepaVit addresses the challenges of sickle cell disease (SCD) detection in resource-limited environments such as sub-Saharan Africa.

Our key contributions are as follows: (1) A dual-branch architecture merging CNN and Transformer features for robust red blood cell (RBC) classification; (2) A residual-normalized fusion mechanism to combine local and global representations effectively; (3) A tailored training pipeline using Focal Loss and data augmentation to mitigate class imbalance; (4) Release of SKDrepaData-v2, a new annotated dataset of blood smear images from African SCD patients; (5) Comprehensive benchmarking against state-of-the-art RBC classification models; (6) Integration of Grad-CAM for interpretability and model transparency; (7) Deployment of an interactive web interface to facilitate clinical adoption.

The remainder of this paper is organized as follows: Section 2 reviews related work on SCD classification. Section 3 details the DrepaVit architecture. Section 4 presents the dataset, experimental setup, and metrics. Section 5 reports and analyses the results. Section 6 discusses comparisons with state-of-the-art methods. Section 7 concludes with future directions.

2 Related Work

Historically, sickle cell disease (SCD) diagnosis has relied on manual microscopy, a process that is time-consuming, expensive, and prone to inter-observer variability [17]. Early AI approaches addressed these limitations through handcrafted feature extraction combined with classifiers like SVM, KNN, or Random Forest [18], [19], [20]. Though applied with some success in RBC classification [21], [22], [23], [24], these methods heavily depend on accurate segmentation and are sensitive to noise and imaging variability.

Deep learning, particularly convolutional neural networks (CNNs), has significantly advanced RBC analysis by learning features directly from images[25]. Xu et al. [14] achieved 87.5% accuracy and 0.94 AUC using a CNN to classify eight RBC types. Other efforts have leveraged U-Net [26], transfer learning[27], and benchmarking with pretrained models [28], [29], [30], [31]. For example, ResNet-50 reached 93.88% accuracy [32], while DAPN-48 achieved 100% [33]. However, CNNs remain limited in modeling long-range dependencies, which are crucial in detecting subtle morphological differences in blood cells[34], [35].

To overcome these limitations, Vision Transformers (ViTs) have emerged, offering strong global contextual modeling [36]. Applications in medical imaging—from mammography [37] to dermatology [38]—have shown ViTs often outperform CNNs, especially in multimodal and fine-grained tasks [34], [39], [40]. Yet, their sensitivity to adversarial noise has prompted robust adaptations like SEViT [41].

Hybrid models that integrate CNNs and Transformers combine local feature extraction with global representation learning. Architectures such as PHTrans [42], CTCNet [43], and MedViT [44] exemplify this synergy. However, in SCD diagnosis, these hybrid models remain underexplored. To bridge this gap, we propose DrepaViT, a novel CNN-Transformer architecture tailored for precise and interpretable classification of sickled red blood cells.

3 Proposed DrepaVit Model

DrepaViT is a hybrid architecture that combines ResNet-50 (CNN)[16] and a Vision Transformer (ViT) [15] to automatically classify red blood cells into three categories—normal, sickled, and other types—from microscopic images, thereby supporting SCD diagnosis. The model consists of two parallel branches that process the input image simultaneously for enhanced robustness:

1. A CNN branch based on ResNet-50 extracts hierarchical local spatial features at multiple depths.
2. A Transformer branch employs ViT to capture global contextual relationships via sequential patch processing.

The input image ($224 \times 224 \times 3$) is fed into both branches. The resulting features are fused, normalized, and passed through a multilayer perceptron (MLP) classifier, which outputs class probabilities for the three target categories. **Fig. 1** illustrates the complete DrepaViT architecture.

3.1 CNN-ResNet-50 Branch

The CNN branch of DrepaViT leverages a ResNet-50 [16] pre-trained on ImageNet as a local feature extractor. To adapt it for our task, we remove the original classification head and retain the convolutional layers up to the final residual block (Block_5), which outputs a feature map $F_{CNN} \in R^{(2048 \times 7 \times 7)}$. A 1×1 convolution is applied for dimensionality reduction, projecting the channels into a latent space of dimension $D_h = 256$ followed by adaptive average pooling and flattening. This yields a compact vector representation $F'_{CNN} \in R^{D_h}$ as formalized in **Eq.(1)**. This vector is subsequently integrated in the fusion module with the global features extracted by the Transformer branch.

$$F'_{CNN} = Flatten \left(AvgPool \left(Conv_{1 \times 1} (F_{CNN}) \right) \right) \in R^{D_h}. \quad (1)$$

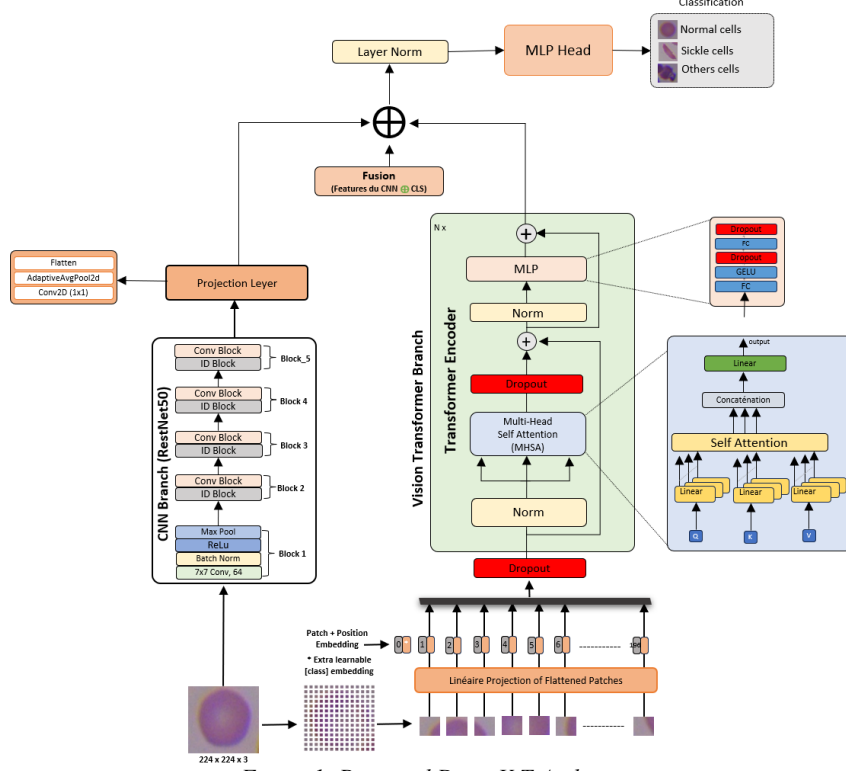


Figure 1: Proposed DrepaViT Architecture

3.2 Transform Branch

The Transformer branch leverages a Vision Transformer (ViT) [15] to capture global dependencies in blood smear images. The input image $X \in R^{3 \times 224 \times 224}$ is divided into $N = (224/16)^2 = 196$ non-overlapping patches of size 16×16 . Each patch is projected into a 256-dimensional latent space using a 2D conv (kernel/stride = 16).

$$x_p = \text{Conv}_{16 \times 16, \text{strides}=16}(X) \in R^{D_h \times \frac{H}{P} \times \frac{W}{P}} \quad (2)$$

The sequence $x_p \in R^{N \times D_h}$ is prepended with a learnable class token $x_{cls} \in R^{1 \times D_h}$, and positional embedding $E_{pos} \in R^{(N+1) \times D_h}$ are added: $x = S[x_{cls}; x_p] + E_{pos}$.

A dropout ($p = 0.1$) is applied after position embedding to regularize learning. The sequence then passes through $L=4$ Transformer blocks, each composed of Multi-Head Self-Attention (MHSA) and a two-layer MLP:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{\frac{D_h}{h}}}\right)V. \quad (3)$$

$$\text{MLP}(x) = \text{Linear}_2\left(\text{GELU}(\text{Linear}_1(x))\right) \quad (4)$$

Each block includes residual connections and LayerNorm following the pre-norm scheme: $x' = x + \text{MHSA}(\text{LayerNorm}(x))$, $x'' = x' + \text{MLP}(\text{LayerNorm}(x'))$

3.3 Fusion and Classification

At the output of the Transformer branch, the class token x_{cls} is fused with the CNN feature vector $F'_{CNN} \in R^{D_h}$ via element-wise addition and LayerNorm: $x_{fused} = \text{LayerNorm}(x_{cls} + F'_{CNN})$.

The fused vector is then passed to a classification head including a linear layer, ReLU activation, dropout, and L2 normalization. The final prediction is computed as:

$$y = \text{Linear}(x_{fused}) \in R^3 \quad (5)$$

This additive fusion effectively combines complementary features: global spatial structure from ViT and fine-grained morphology from CNN, enhancing robustness and accuracy in classifying red blood cells into normal, sickled, and other abnormal types.

4 Experiments

4.1 Dataset: SKDrepaData-v2

We conducted our experiments on SKDrepaData-v2 [45], a publicly available dataset comprising 1,489 Giemsa-stained blood smear images from 20 confirmed SCD patients. Samples were collected from two Senegalese institutions: UMRED (University of Thiès) and the Children’s Hospital of Diamniadio. Images ($224 \times 224 \times 3$) were denoised using Noise2Void [46] and manually annotated by hematologists into three classes: normal, sickled, and other abnormal cells. The dataset reflects clinical class imbalance; distribution details are shown in Fig. 2.



Figure 2: Overview of the dataset showing class distribution and sample cell images

4.2 Class Imbalance and Data Augmentation

Due to the natural imbalance in red blood cell types, two complementary strategies were adopted to improve model generalization:

Data augmentation was applied to the training set using random crops with resizing, horizontal and vertical flips, color jitter, and normalization. These transformations aimed to increase sample diversity, particularly for underrepresented classes, and to reduce overfitting.

Weighted sampling was used during training to balance class frequency across mini-batches by assigning higher weights to minority classes.

The dataset was split into 70% for training (1,131 images), 20% for validation (268 images), and 10% for testing (90 images).

4.3 Training Setup

DrepaViT was trained for 100 epochs using the AdamW optimizer with a CosineAnnealingLR learning rate scheduler. To address class imbalance during training, we used Focal Loss[47], which down-weights the contribution of well-classified samples and focuses learning on hard-to-classify examples. Key hyperparameters are summarized in **Table 1**.

Table 1: Training Hyperparameters

Hyperparameter	Value
Patch size (ViT)	(16 x 16 x 3)
Number of Transformer blocks	4
Hidden dimension (Transformer)	256
MLP dimension (Transformer)	512
Number of attention heads	4
Dropout rate	0.1
Batch size	32
Learning rate	1e-4
Weight decay (L2)	1e-4
Loss function	Focal Loss ($\gamma=2$)
Activation function	GeLU

4.4 Evaluation Metrics

To assess model performance, we used several standard metrics: accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC). AUC was computed from ROC curves by plotting true positive rate against false positive rate across different thresholds. Higher AUC values indicate better class separability.

5 Results and Discussion

5.1 Quantitative Evaluation on SKDrepaData-v2

To assess DrepaViT, we benchmarked it against state-of-the-art CNNs and a baseline Vision Transformer (ViT) on the SKDrepaData-v2 dataset. Results are reported on

training, validation, and test sets. DrepaViT outperformed the ViT model across all splits. On validation, DrepaViT reached 92.54% accuracy vs. 87.50% for ViT (loss: 0.0189 vs. 0.4033). On test data, DrepaViT achieved 93.33% accuracy, 93.82% precision, and 97.56% AUC (**Table 2**), surpassing ViT (accuracy: 80.00%, AUC: 90.43%).

Table 2: Performance of the models on the test set (SKDrepaData-v2)

Model	Acc.	P.	Recall	F1-Score	TNR	MCC	AUC
ViT/16	80	82.57	80	80.23	90	70.84	90.43
DrepaViT	93.33	93.82	93.33	93.30	96.80	90.27	94.56

The training and validation curves indicate that DrepaViT converges faster and more stable than ViT, with lower validation loss and higher accuracy—demonstrating better generalization. These plots are not shown here due to space constraints. Results in **Table 2** and ROC curves in **Fig. 3(a-b)** reveal DrepaViT's superior class separation, particularly on rare or ambiguous cell types. DrepaViT shows minimal confusion between normal, sickled, and other abnormal cells.

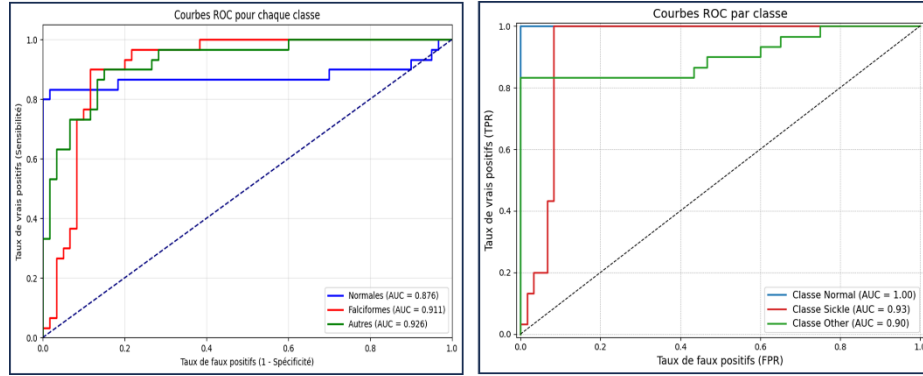


Figure 3 : ROC Curves of the Models Evaluated on the Test Set: (a) ViT, (b) DrepaViT

5.2 Model Interpretability and Feature Analysis

To improve interpretability, we used Grad-CAM[48], which highlights key regions influencing predictions. Although this plot is not shown due to space constraints, DrepaViT consistently localizes discriminative morphological patterns. This results from the complementary strengths of its hybrid CNN–Transformer branches design, combining fine-grained features from ResNet-50 with global context modeling from ViT. Their additive fusion optimally integrates the feature streams.

5.3 Clinical Relevance and Deployment

We developed a web interface [49] enabling clinicians to upload microscopy images and receive real-time cell classification. DrepaViT’s classification accuracies reach

99.97% (normal), 99.76% (sickled), and 90.12% (other), supporting robust deployment in clinical workflows. These results highlight the potential of DrepaViT as a decision-support tool for hematology, particularly in low-resource settings.

5.4 Comparison with State-of-the-Art

DrepaViT significantly outperforms CNN-based baselines such as VGG16/19, MobileNetV2, EfficientNetB0, InceptionV3, and Xception (**Table 3**). While ResNet50 and ViT alone achieve 85.54% and 80.09% respectively, their hybrid fusion yields 93.33% accuracy. Prior works [14], [26], [28], [29], [30] often relied on segmentation or small datasets and lacked robustness. DrepaViT integrates Focal Loss and data augmentation to address class imbalance, achieving superior recall on minority classes (e.g., 93.33% vs. 62.65% for Xception). Grad-CAM analysis confirms DrepaViT's attention to discriminative regions, enhancing both interpretability and trustworthiness.

Table 3: Comparison Results of State-of-the-Art Models on the SKDrepaData-v2 Dataset

Models	Accuracy	Precision	Recall	F1-Score	MCC
MobileNetV2	0.9036	0.9036	0.9036	0.9031	0.8551
EfficientNetB0	0.8675	0.8808	0.8675	0.8708	0.8015
VGG16	0.9036	0.9036	0.9036	0.9036	0.8551
VGG19	0.8434	0.8546	0.8434	0.8394	0.7699
ResNet50	0.8554	0.8607	0.8554	0.8525	0.7831
Xception	0.6265	0.7009	0.6265	0.6077	0.4727
InceptionV3	0.7831	0.7816	0.7831	0.7453	0.6883
ViT/16	0.8000	0.8257	0.8000	0.8023	0.7084
DrepaViT	0.9333	0.9382	0.9333	0.9330	0.9027

6 Conclusion and Future Work

In this study, we introduced DrepaViT, a hybrid deep learning model that integrates a CNN branch (ResNet-50) with a Transformer branch (ViT) for the classification of red blood cells in the context of sickle cell disease diagnosis. By leveraging both local features and global contextual relationships, our approach achieves state-of-the-art performance on the SKDrepaData-v2 dataset, with an accuracy of 93.33%, F1-score of 93.30%, and AUC of 94.56 on the test set.

Interpretability is ensured through Grad-CAM visualizations, highlighting discriminative regions crucial for clinical validation. An interactive web interface further supports its deployment in low-resource settings.

Future work includes hyperparameter tuning and integration of multimodal data to enhance the model's robustness. Overall, DrepaViT offers a promising step toward automated, interpretable, and accessible tool, with strong potential for hematological screening and patient monitoring.

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